Projective Synchronization of Fractional-Order Quaternion-Valued Uncertain Competitive Neural Networks

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Abstract

This article addresses the projective synchronization problem for fractional-order quaternion-valued competitive neural networks with parameter uncertainties. By decomposing the quaternion systems into four real-valued sub-networks and designing robust feedback controllers, projective synchronization criteria are established via fractional Lyapunov stability theory and Linear Matrix Inequalities (LMIs).

1. Introduction

Quaternion-valued neural networks (QVNNs) have gained prominent attention due to their ability to process multi-dimensional data such as 3D computer graphics, color image processing, and robotics trajectory tracking. Competitive neural networks feature two types of state variables: short-term memory (STM) describing neural activity and long-term memory (LTM) describing synaptic modifications.

Projective synchronization refers to synchronization where the master and slave systems synchronize up to a constant scaling matrix alpha = diag(alpha_1, ..., alpha_n). This generalized form includes complete synchronization (alpha = 1) and anti-synchronization (alpha = -1) as special cases.

2. Model Formulation and Quaternion Preliminaries

The skew field of quaternions is denoted by H = {q = q_0 + i q_1 + j q_2 + k q_3 : q_r in R}, where i^2 = j^2 = k^2 = ijk = -1, ij = -ji = k. Due to the non-commutative multiplication in H, direct algebraic treatment is challenging.

Definition 2. *1 (Caputo Fractional Derivative). The Caputo fractional derivative of order alpha in (0, 1) for a continuous function x(t) is defined by ^C D^alpha_t x(t) = (1 / Gamma(1 - alpha)) int_0^t (t - s)^{-alpha} x'(s) ds.*

Assumption 2. *1 (Lipschitz Activation Function). The activation function f_j: H -> H satisfies ||f_j(u) - f_j(v)|| <= L_j ||u - v|| for all u, v in H, with positive constant L_j.*

Lemma 2. *2 (Fractional Quadratic Lyapunov Property). Let x(t) in R^n be a differentiable vector function. Then for any positive definite symmetric matrix P = P^T > 0 and alpha in (0, 1): ^C D^alpha_t [x^T(t) P x(t)] <= 2 x^T(t) P [^C D^alpha_t x(t)].*

3. Main Synchronization Theorems

Consider the master competitive QVNN: ^C D^alpha_t x_i(t) = -c_i x_i(t) + sum_{j=1}^n a_{ij} f_j(x_j(t)) + sum_{j=1}^n b_{ij} v_j(t) + I_i, and the slave QVNN with controller u_i(t). Define synchronization error e_i(t) = y_i(t) - alpha_i x_i(t).

Theorem 3. *1 (Projective Synchronization via Feedback Control). Under Assumption 2.1, if there exist positive constants lambda_i and feedback gain matrices K_i such that the Linear Matrix Inequality [-(c_i + K_i) + L_i ||a_{ij}||] < 0 holds for all i = 1, ..., n, then the slave competitive QVNN achieves global projective synchronization with the master system.*

Theorem 3. *2 (Robust Synchronization with Uncertainties). Consider parameter perturbations Delta a_{ij}(t) bounded by ||Delta a_{ij}(t)|| <= mu_{ij}. Under the robust controller u_i(t) = -K_i e_i(t) - sign(e_i(t)) sum mu_{ij} ||f_j(y_j)||, the uncertain quaternion competitive network achieves projective synchronization asymptotically.*

Remark 3. *3. By projecting quaternion states into four real components (R, I, J, K), the non-commutativity obstacle is resolved without requiring the activation functions to be analytic.*